

GENERAL THETA FOUNDATIONS I: ENTROPY DISSIPATION AND SHARP FINITE-ALPHABET MEMORY

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ABSTRACT. Let a fixed finite family of orthogonal commands have an absolute spectral gap on the full mean-zero L^2 space of the sphere. We prove that its wordwise numerical behavior over N commands requires and admits $\Theta(N^{(d-1)/2})$ persistent classical labels, at zero error and at every fixed sufficiently small output error. The lower bound permits arbitrary hidden states, horizon-dependent design and unrelated rows at different epochs. It removes the logarithmic loss of a packet-mixing argument. The proof compares entropy production under random rotations with the entropy cost of conditional-mean compression after Gaussian smoothing; the latter is charged to one telescoping quadratic loss. A weighted occupation theorem also permits time-dependent command laws and unbounded widths at the other cuts. For five explicit rational three-dimensional commands, the classical width is consequently $\Theta(N)$, or $\log_2 N + O(1)$ persistent label bits, while a fixed qubit realizes the same numerical probabilities. The external spectral-gap input is qualitative, not a numerical gap certificate. Our resource is nonuniform clocked atomic-row width. A separate padded fair-bit compiler has no update-duration side channel and does not identify this width with uniform computational space.

1. INTRODUCTION

A positive factorization at one time does not specify a stochastic machine that remains valid at all later times. This distinction becomes quantitative when a finite register must preserve an orthogonal orbit under every command word. At each isolated cut a fixed-dimensional simplex can encode the observable vector, yet its successive positive realizations need not be compatible. The problem is to determine the number of labels required by one complete causal realization, with all hidden directions allowed.

Fix $U_a \in O(d)$ for a in a finite alphabet $\mathcal{A}$, seeds $x \in X = \{\pm e_1, \dots, \pm e_d\}$ and signal $0 < \rho \leq 1/10$. The last command acts on the left: $U_w = U_{a_N} \cdots U_{a_1}$. A query $j \in \{1, \dots, d\}$ is supplied only after the N commands. The required binary response is

$$(1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho e_j^T U_w x}{2}, \quad b \in \{-1, 1\}.$$

All probabilities lie in $[9/20, 11/20]$. Equality, or the stated uniform error, is required on every word, not merely under a stationary input law. Let $W_{N,\varepsilon}$ be the least maximum number of persistent labels with binary row total-variation error at most ε ; the complete machine convention is Definition 2.1.

For a probability vector p on $\mathcal{A}$, write

$$(2) \quad Pf(v) = \sum_a p(a) f(U_a^{-1}v), \quad \|P|_{L_0^2(\sigma)}\| \leq \lambda < 1,$$

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where σ is normalized surface measure on $\mathbb{S}^{d-1}$. The inverse convention makes P also the density transfer operator. The same norm condition holds for its adjoint. It is a hypothesis on *all* spherical harmonics, not the spectrum of the defining d -dimensional representation.

Theorem 1.1 (Sharp finite-alphabet width). *Suppose (2) holds and $0 \leq \varepsilon < \rho/(2\sqrt{d})$. Then, for constants depending only on $d, \lambda, \rho, \varepsilon$,*

$$(3) \quad cN^{(d-1)/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq C_d N^{(d-1)/2} \quad (N \geq 1).$$

The lower bound allows arbitrary hidden state continuations and cut-dependent stochastic rows. The upper realization has common command rows and an N -dependent terminal decoder. For rational commands, seeds and signal, its rows can all be rational and each update has at most $d+1$ successors. If the identity is available and $d\rho < 1$, every isolated cut nevertheless has positive minimum $d+1$.

The preceding finite-alphabet result [14] gave lower order $(N/\log(N+1))^{(d-1)/2}$ and the same upper construction. Its converse waited $O(\log k)$ command slots for the product distribution to become uniform to the resolution of a k -center quantizer. The proof here does not shorten that mixing time or assert one-step total-variation mixing of an atomic walk. It retains an entropy potential across *all* updates, so that repeated mixing need not be purchased afresh.

Put $Y_t = U_{a_t} \cdots U_{a_1} x$ and $Z_t = \mathbb{E}[Y_t \mid S_t]$. The auxiliary Gaussian density $f_t = \text{Law}(Z_t) * \gamma_\tau$ is used only in the proof. Randomizing a command increases its entropy by an angular Hellinger variance times the spectral gap. Replacing the resulting vector by a conditional mean decreases entropy by at most

$$\frac{\mathbb{E}\|Z_t\|^2 - \mathbb{E}\|Z_{t+1}\|^2}{2\tau^2}.$$

This cost telescopes. If Z_t has only k possible values but has nonvanishing mean norm, smoothing at scale $\tau \asymp k^{-1/(d-1)}$ still leaves a definite angular Hellinger variance. These statements combine to give an $O(k^{2/(d-1)})$ occupation bound, with no logarithmic factor. No noise is added to the actual command experiment.

The five rational gates of [14] consequently have exact and fixed-error classical width $\Theta(N)$. Their one-qubit realization is unchanged. This is a separation in worst-case numerical behavior under externally supplied words: the classical label bits are $\log_2 N + O(1)$, not linear bit space. The gap for these gates is supplied by Bourgain–Gamburd [1]; we do not give a numerical gap value.

Quantity	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at a cut; smallest whole-run peak.
q_t, v_t	$\mathbb{E}\ Z_t\ $ and $\mathbb{E}\ Z_t\ ^2$, proof-side conditional moments.
$h(f), \Pi$	Differential entropy with natural logarithms; spherical averaging at each radius.
$g_t = 1 - \lambda_t^2$	The squared-norm gap of the command law at epoch t .
κ	A positive lower bound on retained terminal amplitude.

The new results are the one-step entropy comparison, the gap-weighted occupation law and its sharp width consequences. Rational spherical synthesis and the qubit operations are inherited and reproved. Spectral gaps, Gaussian score identities, Hellinger comparison and spherical quantization are classical inputs. Appendix A repairs the implementation timing convention; it is not counted as a new sampling algorithm. Section 7 gives the corresponding theorem comparisons. The prior packet, monomial and cyclic results remain intact in the supporting article rather than being silently weakened or used under altered hypotheses.

2. THE EXPERIMENT AND ITS REGISTERS

Definition 2.1 (Clocked stochastic realization). For a prescribed horizon N , a machine consists of finite alphabets $[K_t]$, $0 \leq t \leq N$, initial distributions E_x , row-stochastic matrices $T_{t,a}$ of size $K_t \times K_{t+1}$, and columns $d_j \in [-1, 1]^{K_N}$. Its response mean is $E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j$. Its error is one half the maximum absolute discrepancy from $\rho e_j^T U_w x$ over the finite set of seeds, words and queries. Its peak is $\max_t K_t$, also including two labels if the final binary answer is retained at a charged cut. The minimum peak at error ε is $W_{N,\varepsilon}$.

The seed and query are each supplied once. The current command is consumed in its update. Only the updated register survives; a retained seed, input transcript, random seed, or data-dependent scheduling variable would be part of that register. Available, not merely reachable, labels are counted; padding therefore gives a larger admissible profile. Every unused row is normalized. The epoch, horizon, row tables, their construction, lookup and arithmetic, and exact atomic stochastic sampling are free. A machine may be redesigned for every N and need not use common rows. This is a nonuniform probabilistic ordered read-once width, not total program size, autonomous state count or uniform computational space.

All converse distributions below are imposed on the external word, independently of private randomness. Wordwise correctness licenses this choice; the test law is not additional runtime information. The commands in this paper's new converse are independent, with a possibly different law at each epoch. No independence of the hidden past given its current state is assumed.

Lemma 2.2 (Conditional moments and calibration). *Fix a unit seed x . Let the fresh command a_{t+1} be independent of the past and let private updates obey the stated Markov rule. Define*

$$Z_t = \mathbb{E}[Y_t | S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2, \quad Y_t = U_{a_t} \cdots U_{a_1} x.$$

Then $\|Z_t\| \leq 1$, q_t and v_t are nonincreasing, and

$$(4) \quad Z_{t+1} = \mathbb{E}[U_{a_{t+1}} Z_t | S_{t+1}], \quad \Delta_t := v_t - v_{t+1} = \mathbb{E}\|U_{a_{t+1}} Z_t - Z_{t+1}\|^2.$$

In particular $\sum_{t < N} \Delta_t \leq 1$. If the machine has wordwise error at most ε , then

$$(5) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{d} - 2\varepsilon.$$

Proof. Conditioning on S_t, a_{t+1} and its randomized successor does not change the conditional mean of Y_t : the new command is independent of the past and the update uses that past only through S_t . The tower property gives (4). Conditional Jensen and the orthogonality of U_a give both monotonicities. The orthogonal projection identity for conditional expectation gives the equality for Δ_t . Initially $Z_0 = x$, even when initialization randomizes the register.

In the proof only, combine the separate decoder columns as

$$D(S_N) = (d_1(S_N), \dots, d_d(S_N)), \quad \|D(S_N)\| \leq \sqrt{d}.$$

For every word its conditional mean is within $2\sqrt{d}\varepsilon$ of ρY_N . Hence

$$\rho - 2\sqrt{d}\varepsilon \leq \mathbb{E}\langle Y_N, D(S_N) \rangle = \mathbb{E}\langle Z_N, D(S_N) \rangle \leq \sqrt{d}q_N.$$

This does not require jointly sampling multiple answers. The vector combines different permitted final query columns only in the proof. $\square$

Proposition 2.3 (The isolated-cut minimum). *If $\text{Id} \in \mathcal{A}$ and $d\rho < 1$, the normalized residual rank and separate positive minimum at every internal cut equal $d + 1$.*

Proof. The residuals are affine functions of $\rho U_w x$. Idle prefixes from $\pm e_j$ span $\mathbb{R}^d$, and idle suffixes with coordinate queries separate these coordinates. Normalization adds an independent constant direction, so the rank is $d + 1$. A regular simplex with inradius ρ has circumradius $d\rho < 1$ and

contains every reachable predictive vector. Its vertices therefore define legal binary response channels after every orthogonal suffix. Decompose the selected-cut vector into those vertices. One may store the raw prefix before this cut and the raw suffix together with the simplex label after it. Those other state sets are finite and uncapped. Thus this is a factorization within a complete machine. Rank gives the matching lower bound. This proposition, inherited from [14], does not assert simultaneous attainment. $\square$

3. ENTROPY COST OF A CONDITIONAL CENTROID

For $\tau > 0$, let γ_τ be the centered Gaussian density on $\mathbb{R}^d$ with covariance $\tau^2 I$. If X is bounded, write $f_X = \text{Law}(X) * \gamma_\tau$ and $h(f) = -\int_{\mathbb{R}^d} f \log f$. All entropies and divergences in this section use natural logarithms. Gaussian mixtures with bounded centers have finite entropy and the Gaussian tail bounds needed below.

Lemma 3.1 (Quadratic cost of barycentric compression). *Let X be a bounded random vector and J a finite random variable on the same probability space. Set $M = \mathbb{E}[X | J]$. Then*

$$(6) \quad h(f_X) - h(f_M) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

No sign assertion for the left side is required.

Proof. Put $f = f_M$. Direct differentiation of the finite Gaussian mixture gives

$$(7) \quad \nabla^2 \log f(y) = -\tau^{-2} I + \tau^{-4} \text{Cov}(M | M + \tau G = y) \succeq -\tau^{-2} I,$$

where G is an independent standard Gaussian. This identity also follows by differentiating the log-sum of exponentials after factoring out $\exp(-\|y\|^2/(2\tau^2))$. Define $\varphi(x) = -\mathbb{E}_G \log f(x + \tau G)$. Boundedness of the centers justifies differentiation and gives $\nabla^2 \varphi \preceq \tau^{-2} I$. Therefore

$$\varphi(X) - \varphi(M) \leq \langle \nabla \varphi(M), X - M \rangle + \frac{\|X - M\|^2}{2\tau^2}.$$

The first term averages to zero conditional on J . On the other hand, nonnegativity of $D(f_X \| f)$ gives

$$h(f_X) - h(f) \leq -\int (f_X - f) \log f = \mathbb{E}\varphi(X) - \mathbb{E}\varphi(M).$$

These inequalities prove the result. The Gaussian density used for convolution is a common mathematical kernel; no additional noisy state is assumed in the transducer. $\square$

The Gaussian score/Hessian mechanism is classical. For comparison, Polyanskiy–Wu [2, Proposition 1 and Corollary 4] control smoothed entropy differences by a Wasserstein distance. Here the specified martingale coupling $M = \mathbb{E}[X | J]$ cancels the linear Taylor term. The resulting one-sided *squared* error, rather than a square-root continuity estimate for unrelated measures, is what can be summed along a stochastic register. We use the explicit proof above, not an assumed entropy monotonicity under convex order.

3.1. An entropy increment from a spherical gap. Let Π be spherical averaging at each radius on $L^2(\mathbb{R}^d)$:

$$(\Pi u)(r\omega) = \int_{\mathbb{S}^{d-1}} u(r\theta) d\sigma(\theta).$$

It is the orthogonal projection onto radial functions. The angular gap (2), integrated in polar coordinates, gives

$$(8) \quad \|Pu\|_2^2 \leq \|\Pi u\|_2^2 + \lambda^2 \|u - \Pi u\|_2^2.$$

Here P acts on Euclidean functions by the same orthogonal pullbacks; it preserves radial functions and commutes with Π .

Lemma 3.2 (Entropy increment and angular Hellinger variance). *For a Gaussian mixture f with bounded centers,*

$$(9) \quad \mathbf{h}(Pf) - \mathbf{h}(f) \geq (1 - \lambda^2) \|\sqrt{f} - \Pi\sqrt{f}\|_2^2.$$

Without a strict gap the entropy increment is still nonnegative.

Proof. Set $f_a(y) = f(U_a^{-1}y)$ and $g = \sum_a p(a)f_a = Pf$. All f_a have the same entropy, and hence

$$\mathbf{h}(g) - \mathbf{h}(f) = \sum_a p(a) D(f_a \| g).$$

For probability densities u, v , Jensen's inequality gives $D(u \| v) \geq -2 \log \int \sqrt{uv} \geq 2(1 - \int \sqrt{uv}) = \|\sqrt{u} - \sqrt{v}\|_2^2$. Consequently the entropy increment is at least

$$\begin{aligned} \sum_a p(a) \|\sqrt{f_a} - \sqrt{g}\|_2^2 &= 1 - \|P\sqrt{f}\|_2^2 + \|P\sqrt{f} - \sqrt{g}\|_2^2 \\ &\geq 1 - \|P\sqrt{f}\|_2^2. \end{aligned}$$

As $\|\sqrt{f}\|_2^2 = 1$, (8) proves the assertion. The last statement also follows directly from entropy concavity. This argument uses the full L^2 gap only on square roots of smoothed probability densities; it assumes no logarithmic Sobolev inequality and no one-step mixing of point masses. $\square$

4. SPARSE CENTROIDS CANNOT BE ANGULARLY UNIFORM

Write $m = d - 1$. Fix a cap constant $A_d \geq 1$ such that

$$(10) \quad \sigma\{\theta : \|\theta - u\| \leq r\} \leq A_d r^m \quad (0 < r \leq 1, u \in \mathbb{S}^{d-1}).$$

For example, with $|\mathbb{S}^j|$ denoting surface area, one can use

$$A_d = \max \left\{ 1, \frac{|\mathbb{S}^{m-1}|}{m|\mathbb{S}^m|} \left(\frac{\pi}{2} \right)^m \right\}.$$

Indeed a chordal radius r has angular radius $2 \arcsin(r/2) \leq \pi r/2$, and polar integration with $\sin t \leq t$ proves the estimate. In dimension three, $A_3 = 1$ is a convenient nonoptimal choice, since the exact cap mass is $r^2/4$.

Lemma 4.1 (A scale-uniform sparse-mixture obstruction). *Suppose $\|Z\| \leq 1$, Z has at most k values, and $\mathbb{E}\|Z\| \geq \kappa$ for $0 < \kappa \leq 1$. Put*

$$(11) \quad r = \left(\frac{\kappa}{16A_d k} \right)^{1/m}, \quad \tau = \frac{\kappa r}{8\sqrt{2d}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then

$$(12) \quad \|\sqrt{f} - \Pi\sqrt{f}\|_2^2 \geq \frac{\kappa}{16}.$$

The locations and masses of the atoms are arbitrary.

Proof. Let $L = \{\|Z\| \geq \kappa/2\}$. Since $\mathbb{E}\|Z\| \leq \kappa/2 + \mathbb{P}(L)$, its probability is at least $\kappa/2$. Around each direction of a nonzero atom satisfying L place a spherical cap of chordal radius r . Their union Ω satisfies $\sigma(\Omega) \leq kA_d r^m = \kappa/16$.

Let G be independent standard Gaussian. Markov's inequality gives $\mathbb{P}(\|G\| \leq \sqrt{2d}) \geq 1/2$. On this event and L ,

$$\|\tau G\| \leq \kappa r/8, \quad \left\| \frac{Z + \tau G}{\|Z + \tau G\|} - \frac{Z}{\|Z\|} \right\| \leq \frac{2\|\tau G\|}{\|Z\|} \leq r/2.$$

The denominator is nonzero here because $r \leq 1$ and $\|\tau G\| \leq \|Z\|/4$. Thus the radial cone $C_\Omega = \{s\theta : s > 0, \theta \in \Omega\}$ has $\int_{C_\Omega} f \geq \kappa/4$.

Write $u = \Pi\sqrt{f}$. This is radial, and $\|u\|_2 \leq 1$. Polar integration yields $\int_{C_\Omega} u^2 = \sigma(\Omega)\|u\|_2^2 \leq \kappa/16$. The triangle inequality in $L^2(C_\Omega)$ now gives

$$\|\sqrt{f} - u\|_2 \geq \sqrt{\kappa/4} - \sqrt{\kappa/16} = \frac{\sqrt{\kappa}}{4}.$$

Squaring proves the assertion. Repeated atom directions and zero atoms only reduce the number of caps. No regularity or separation assumption on the centroid locations was used. $\square$

The cap scale in this lemma is the classical $k^{-1/m}$ quantization scale. To fix the comparison precisely, for a sphere probability measure ν its k -center cosine distortion is

$$\inf_{u_1, \dots, u_k \in \mathbb{S}^m} \int \min_i (1 - \langle v, u_i \rangle) d\nu(v) = \frac{1}{2} \inf_{u_1, \dots, u_k \in \mathbb{S}^m} \int \min_i \|v - u_i\|^2 d\nu(v).$$

Repeated centers are permitted. For surface measure its order is $k^{-2/m}$; the cap lower estimate and a covering upper estimate already prove that order. This is part of classical quadratic quantization, not a new geometric exponent; see Graf–Luschgy [3] and the manifold extension of Iacobelli [4, Theorem 1.4]. The quantity in (12) is different: it is an angular Hellinger obstruction for a *Gaussian-smoothed* finite register. Combined with the one-step entropy increment, it need not wait for a word product to approximate Haar measure at this scale.

5. A SHARP OCCUPATION LAW

The argument is useful in an inhomogeneous form. At epoch t , let the fresh command have a prescribed law p_t , independently across epochs, and let its angular operator satisfy

$$\|P_t|_{L_0^2(\sigma)}\| \leq \lambda_t \leq 1, \quad g_t = 1 - \lambda_t^2.$$

The alphabets and orthogonal matrices can also depend on t in this probabilistic statement. No strict gap is required at every epoch. Define the dimension-only constants

$$(13) \quad a_d = 3 + \frac{2}{d-1}, \quad B_d = 2048d(16A_d)^{2/(d-1)}.$$

Theorem 5.1 (Gap-weighted occupation without a mixing-time loss). *For the hidden process of Lemma 2.2, suppose $q_N \geq \kappa > 0$. For every integer $k \geq 1$,*

$$(14) \quad \sum_{\substack{0 \leq t < N \\ K_t \leq k}} g_t \leq B_d \kappa^{-a_d} k^{2/(d-1)}.$$

The other cuts may have arbitrarily many labels. The inequality also holds when K_t is replaced by the number of positive-probability states under this independent-command test law.

Proof. Fix k and choose one smoothing scale τ by (11); keep it fixed throughout the run. Set $f_t = \text{Law}(Z_t) * \gamma_\tau$ and $\tilde{Z}_t = U_{a_{t+1}} Z_t$. Orthogonal invariance of Gaussian noise and independence of the fresh command imply $f_{\tilde{Z}_t} = P_t f_t$. By (4), $Z_{t+1} = \mathbb{E}[\tilde{Z}_t | S_{t+1}]$. Lemma 3.1 therefore yields

$$(15) \quad h(f_{t+1}) - h(f_t) \geq h(P_t f_t) - h(f_t) - \frac{\Delta_t}{2\tau^2}.$$

Whenever $K_t \leq k$, Lemmas 3.2 and 4.1, together with $q_t \geq q_N \geq \kappa$, bound the entropy increment from randomizing the command below by $g_t \kappa/16$. At all other cuts that increment remains nonnegative.

Sum (15). The total quadratic loss is at most one. Furthermore, entropy concavity gives $h(f_0) \geq \frac{d}{2} \log(2\pi e \tau^2)$, while $\mathbb{E}\|Z_N + \tau G\|^2 \leq 1 + d\tau^2$ and Gaussian maximal entropy give

$$h(f_N) \leq \frac{d}{2} \log(2\pi e(\tau^2 + 1/d)).$$

The latter bound follows directly by comparing f_N with the centered Gaussian of that covariance and using nonnegative relative entropy. It does not require Z_N to be centered. Consequently

$$(16) \quad \frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{d}{2} \log \left(1 + \frac{1}{d\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Since $\tau^{-2} = 128d\kappa^{-2}(16A_d k/\kappa)^{2/(d-1)}$, this is (14). Nothing in this summation restricts the intermediate hidden affine dimension, projected vertices, or the widths of the other registers. Under the test law, only positive-mass labels occur in the mixture f_t , proving the final assertion. $\square$

For a fixed gap $g = 1 - \lambda^2 > 0$, (14) gives $\#\{0 \leq t < N : K_t \leq k\} \leq B_d g^{-1} \kappa^{-a_d} k^{2/(d-1)}$. Counting endpoints $1, \dots, N$ instead changes this bound by at most one. Thus the earlier logarithmic loss in occupation, as well as in peak width, is removed. The proof does not assume a spectral gap for the stochastic state matrices; its gap belongs to the *external* orthogonal action on the sphere.

Lemma 5.2 (Exact spherical synthesis, inherited upper bound). *For every $N \geq 1$ and finite orthogonal command alphabet, the experiment (1) admits a common-row realization with at most $C_d N^{(d-1)/2}$ labels. Rational commands, seeds and signal admit rational rows with at most $d+1$ successors.*

Proof. This is the construction from [14]; we include it to make the matched theorem self-contained. Write $m = d - 1$ and $\eta = 1 - 1/(8N)$. The two rational stereographic charts

$$t \mapsto \left(\frac{2t}{1 + \|t\|^2}, \pm \frac{1 - \|t\|^2}{1 + \|t\|^2} \right), \quad t \in [-1, 1]^m,$$

are 2-Lipschitz. Every sphere point has a preimage in the unit ball in one chart. On a grid of mesh $1/M$, with $M = \lceil 2\sqrt{mN} \rceil$, their images form a rational set V of at most $2(2M+1)^m$ points. Every unit direction is within chordal distance $\sqrt{m}/M$ of V . Hence $\eta\mathbb{B}^d \subset P := \text{conv } V$ by the support-function criterion.

Associate label $u \in V$ at time t with $a_t u$, where $a_t = \eta^{-t}/4$. On command a choose a convex representation of $\eta U_a u$ in V . These are common command rows. Initialize the mean to ρx in $a_0 P$, which is possible because $\eta/4 \geq 7/32 > \rho$. Use terminal mean $a_N e_j^T u$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7 < 1$. The identity $a_{t+1} \mathbb{E}[u' \mid u, a] = a_t U_a u$ proves exactness on every word. For rational data, each convex feasibility problem has a rational basic solution with at most $d+1$ nonzero coefficients. The state holds only the label. A charged final bit uses two labels and is absorbed in C_d . No efficient table-generation algorithm is inferred from finite feasibility. $\square$

Proof of Theorem 1.1. Use the independent command law p from (2) and any fixed seed. Lemma 2.2 gives $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. With $K = \max_t K_t$, Theorem 5.1 gives the explicit bound

$$(17) \quad K \geq \left(\frac{(1 - \lambda^2) \kappa^{a_d}}{B_d} N \right)^{(d-1)/2}.$$

The preceding lemma proves the exact upper bound. Proposition 2.3 proves the isolated-cut assertion. $\square$

Corollary 5.3 (An obstruction for actual terminal error). *For an available-label profile $\mathbf{K}$, let $\mathcal{E}_N(\mathbf{K})$ be the least actual worst-row terminal error among all machines with that profile. For every integer k with $I_k := \#\{0 \leq t < N : K_t \leq k\} > 0$,*

$$(18) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{d}} - \left(\frac{B_d k^{2/(d-1)}}{(1 - \lambda^2) I_k} \right)^{1/a_d} \right]_+.$$

In particular, peak $o(N^{(d-1)/2})$ entails $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{d})$.

Proof. The finite parameter domain of a fixed-profile machine is compact and the wordwise objective continuous, so the minimum exists. Apply (14) with $\kappa = q_N$ when $q_N > 0$, and use $q_N \geq \rho/\sqrt{d} - 2\mathcal{E}_N$. If $q_N = 0$, the same conclusion follows directly from calibration. The last assertion uses $I_K = N$. $\square$

This is an actual-error bound: it does not add local defects or rule out their cancellation. If a separately supplied diagram has an accumulated local deficiency certificate D , the classical composition argument produces a machine with final error at most D ; only then does (18) lower-bound that certificate. It need not equal the minimum terminal error.

6. A SHARP RATIONAL EXAMPLE AND ITS QUANTUM REALIZATION

Use the fixed five-letter alphabet $\mathcal{A}_* = \{\text{Id}, R_x, R_x^{-1}, R_z, R_z^{-1}\}$, where

$$(19) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -7/25 & -24/25 \\ 0 & 24/25 & -7/25 \end{pmatrix}, \quad R_z = \begin{pmatrix} -7/25 & -24/25 & 0 \\ 24/25 & -7/25 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

These are the rational commands from [14]. Take $\rho = 1/10$, six seeds $\{\pm e_j : 1 \leq j \leq 3\}$ and three final coordinate queries. No alphabet grows with the horizon.

Theorem 6.1 (Linear classical label complexity). *For every fixed $0 \leq \varepsilon < 1/(20\sqrt{3})$,*

$$(20) \quad W_{N,\varepsilon}(\mathcal{A}_*) = \Theta_\varepsilon(N), \quad \lceil \log_2 W_{N,\varepsilon}(\mathcal{A}_*) \rceil = \log_2 N + O_\varepsilon(1).$$

The exact upper machine may have rational common command rows. The separate positive minimum at each cut is four. If $\lambda_ < 1$ is the norm-gap constant for the walk with probabilities $(1/2, 1/8, 1/8, 1/8, 1/8)$, every such machine has*

$$(21) \quad K \geq \frac{(1 - \lambda_*^2)(1/(10\sqrt{3}) - 2\varepsilon)^4}{98304} N.$$

The existence of $\lambda_ < 1$ is an external spectral-gap input; its numerical value is not evaluated here.*

Proof. Let $\sigma_x, \sigma_y, \sigma_z$ be the Pauli matrices and define $A_x = (3I - 4i\sigma_x)/5$, $A_z = (3I - 4i\sigma_z)/5$ in $\text{SU}(2)$. Their adjoint actions are (19). Their eigenphases include $(3 + 4i)/5$, which is not a root of unity: its sum with its inverse is $6/5$, not an integer, whereas a rational algebraic integer is an integer. The closures of the respective cyclic subgroups therefore contain the two axial circles. The Lie algebra generated by $i\sigma_x, i\sigma_z$ also contains their bracket, a nonzero multiple of $i\sigma_y$. The closed generated subgroup is $\text{SU}(2)$.

The generators have algebraic entries. Bourgain–Gamburd [1, Theorem 1] supplies the upper spectral gap for their symmetric averaging operator on $L^2(\text{SU}(2))$. Adding weight $1/2$ at the identity makes its spectrum nonnegative and its mean-zero norm strictly less than one. Under the left action, lift a sphere function by $f(v) \mapsto f(\text{Ad}_g e_3)$. This is an isometry from the homogeneous quotient $L^2(\mathbb{S}^2)$ into $L^2(\text{SU}(2))$ and intertwines the left pullbacks. Thus the sphere operator has the same norm bound; inverse conventions are harmless for this symmetric walk.

Apply Theorem 1.1 and use $A_3 = 1$ in (13), giving $a_3 = 4$ and $B_3 = 98304$. This proves (21) and the matching upper order. Rational sparse rows follow from Lemma 5.2, and Proposition 2.3 gives four. Taking binary logarithms proves the bit statement, including the integer ceiling. $\square$

The rationality and alphabet are explicit, but a numerical finite-horizon lower threshold requires an additional certified bound on λ_* . We provide neither such a certificate nor a claim that finite-dimensional matrix eigenvalues establish the infinite-dimensional gap. Formula (21) gives the full dependence on any supplied gap certificate. The new conclusion here is removal of the logarithmic factor from the classical label count, not a new construction of expanding generators.

Proposition 6.2 (One qubit with reset, unitary commands and final query). *One qubit realizes (1) for (19) exactly, using a fixed finite collection of operations independent of N . Thus its quantum memory dimension is two, whereas its classical atomic label complexity and label bits satisfy (20) even at the stated fixed positive error. This is a numerical-behavior comparison, not a strict-cutpoint language-state separation.*

Proof. On seed x reset the qubit to $\varrho_x = (I + \rho \sum_j x_j \sigma_j)/2$. Its eigenvalues are positive because $\rho < 1$. Commands apply the unitaries $I, A_x^{\pm 1}, A_z^{\pm 1}$. The Bloch vector after the word is $\rho U_w x$. Measurement of σ_j has outcome b with probability $(1 + b \rho e_j^T U_w x)/2$. Only one final query is made. A seed reset is a completely positive operation; it is not being called a unitary measure-once transition. The classical statement is Theorem 6.1. $\square$

The same qubit operations appeared in the preceding revision. What is strengthened is the matching classical cost. In particular this is one qubit versus $\log_2 N + O(1)$ classical *label bits*, not one qubit versus linear bit space or an entropy lower bound for a chosen stationary input distribution.

7. COMPARISONS AND SCOPE

7.1. Quantization, smoothing and the new accumulated estimate. Classical quadratic quantization minimizes the expected squared distance to a finite codebook. The cosine loss on a sphere is exactly one half of the squared chordal distance. Graf–Luschgy [3] develops the general quantization framework, and Iacobelli [4, Theorem 1.4] proves the corresponding manifold asymptotics under explicit integrability hypotheses. On the compact sphere these give the familiar exponent $2/(d-1)$. Neither this geometric exponent nor the cap argument used here is a priority claim. Our lower bound concerns a sequence of *hidden conditional centroids*, not independently selected nearest neighbors in a static codebook.

Gaussian smoothing and entropy continuity are also established tools. The bound in [2, Corollary 4] applies to unrelated smoothed laws and is linear in an appropriate Wasserstein distance. Lemma 3.1 uses the particular conditional-mean coupling and a semiconvex Gaussian log density to remove the first-order term. We do not assert that this elementary Gaussian identity is absent from the literature. Its role here is to make the entire cost bounded by one quadratic variation budget, rather than by the sum of square roots of individual costs.

The entropy increment in Lemma 3.2 uses the standard relative-entropy/Hellinger comparison and an L^2 spectral bound, not a logarithmic Sobolev assumption. Lemma 4.1 gives a uniform positive angular defect after smoothing at the register’s resolution. The conclusion that combines these ingredients is Theorem 5.1: a finite horizon with arbitrarily varying hidden implementations has a gap-weighted small-register occupation of order at most $k^{2/(d-1)}$. That inequality, rather than a new quantization exponent or expansion theorem, is the main assertion.

7.2. Positive realization and branching programs. After padding layer alphabets, the model is a probabilistic ordered read-once layered program, with command-dependent row-stochastic matrices and a bounded real output. The scalar decoder for one query can be turned into randomized Boolean acceptance by one Bernoulli row. The total number of layer nodes and transition-table size are not the maximum width. In particular the result is not a lower bound in a uniform algorithmic-space model.

Fourier-growth and pseudorandomness results for regular, permutation and width-three branching programs [10, 11, 12] use their stated transition assumptions and control Boolean-input Fourier coefficients. Their regularity is not implied by row stochasticity: a reset matrix with every row $(1, 0, \dots, 0)$ does not preserve the uniform state law. Some bounds for unrestricted deterministic tables do extend to random tables by convexity, so randomness alone is not an

originality boundary. Here the assumed gap is on the external sphere action, whereas the arbitrary hidden matrices need have no gap or regularity. The converse uses the entropy of a smoothed conditional orbit mean, not an estimate on the Fourier mass of a Boolean acceptance function. We make no claim of an exhaustive branching-program priority classification.

Earlier positive-realization mechanisms and the packet theorem remain useful [13, 14]. In particular the full-word endpoint kernel must be allowed to distinguish different spellings of the same product. The present one-step proof uses actual independent command letters, not a product-sufficiency reduction. No previously corrected packet quantifier is relaxed.

7.3. Which quantum memory is compared. Quantum advantages for process simulation precede this work. Gu et al. [5] construct models with smaller stored entropy for classical stochastic processes. Thompson et al. [6] also treat *input-output* processes, not only autonomous outputs: their framework includes stationary input-dependent memory entropy and its worst-input counterpart. Our claim must not be distinguished from that work merely by the presence of input commands. The distinction here is finite-horizon worst-word numerical error, arbitrary horizon-specific classical tables, and maximum available label count; we do not minimize stationary memory entropy.

Ghafari et al. [7] demonstrate a memory-dimension advantage for stochastic process simulation. Thus dimension, as well as entropy, has established quantum-memory comparisons. Proposition 6.2 is an explicit driven family whose *classical* cardinality has a matched growth law in the horizon at fixed positive error. It is not a first claim of quantum dimension advantage. The recent world-model comparison [9] is also relevant; its process and update requirements are not replaced by our nonuniform one-query convention.

For strict cutpoints, Chen–Wu [8, Theorem 3.2] gives a classical PFA with at most $n^2 + 1$ states for a general n -dimensional one-way quantum automaton. The conversion preserves a threshold sign while allowing a length-dependent shrinking numerical margin [8, Proposition 3.1]. It therefore does not preserve the fixed terminal correlation used here. The result in (20) does not contradict or improve that theorem in its own language metric.

7.4. Conclusions. The full-sphere spectral-gap class has matched exact and fixed-error width $\Theta(N^{(d-1)/2})$. The explicit three-dimensional gate family has linear classical label count and logarithmic classical label bits. The proof allows hidden directions, cut-dependent rows, and fixed positive error, and it does not repeat a mixing-time charge.

The constants in the occupation theorem are not optimized. Neither an exact leading asymptotic constant nor every finite integer optimum is identified. A numerical lower constant for the displayed gates still requires a certified spectral-gap value. Arbitrary compact actions without the assumed sphere gap, all irrational cyclic arithmetic, multiple jointly constrained queries and adaptive experimental design are different problems. The retained monomial and cyclic theorems keep their own hypotheses in the supporting article.

Finally, nonuniform atomic-row width and its padded finite-coin implementation are different resources. The latter has the additional control factor proved below. Both leave table construction and storage outside the counted register, and neither is represented as a uniform space bound. The complete predecessor development is preserved separately; it is not needed to validate the new proof or part of the editorial claim for this article.

APPENDIX A. FIXED-DURATION FINITE-COIN IMPLEMENTATION

We make the timing repair explicit rather than assuming that duration is unobservable. The environment supplies a command at a ready boundary. Every completed update uses the same publicly specified number of microsteps, independent of the private bits, old label and command. No output or internal decision is visible during the update; only its fixed-time ready boundary is

visible. Threshold tables are nonuniform read-only data. The bit position, input latch, old label and control variables are counted. There is no uncharged microstep clock.

Theorem A.1 (Padded sparse-row compiler). *Let a finite transducer have peak K , common command rows and at most $L \geq 2$ successors per initialization, command or decoder row, with fixed seed, command and query alphabets. For $\delta > 0$, put*

$$b = \max \left\{ 1, \left\lceil \log_2 \frac{(N+2)(L-1)}{\delta} \right\rceil \right\}, \quad D = (L-1)b.$$

It admits a fair-bit implementation with exactly D private fair-bit microsteps per macro update, terminal error increased by at most δ , and at most $C_{A,X,J,L}K(b+1)$ persistent labels. Its duration discloses no information about the private trajectory.

Proof. For each categorical row, consider its conditional binary decision tree, with at most $L-1$ internal nodes. Replace each conditional probability by a lower dyadic approximation with b bits, keeping probabilities zero and one exact. Coupling decisions bounds the row total variation by $(L-1)2^{-b}$. Zero-mass nodes are given arbitrary legal rows.

To draw a threshold $m/2^b$, compare b private fair bits with its stored binary digits. Until the first mismatch the sampled prefix is equal to the table prefix, so no prefix integer is needed. Store the bit position and the relation (equal, below or above); after a mismatch, continue consuming the remaining bits of this comparison without changing the relation. After b steps, move to the selected child of the categorical tree. Once a leaf has been selected, hold its new label and continue consuming dummy bits until exactly D steps have elapsed. Deterministic thresholds and trees of shorter depth are padded in the same way.

Active states retain the old label, current input, tree node, relation and counted progress; their number is at most a constant times $K(b+1)$ for fixed alphabets and L . After leaf selection the old label is released; the sampler holds only the new label and counted progress. It does not hold two independent K -valued labels. Ready, initialization and query phases add only the same order. A final binary answer can be held inside the padding states. All timing information is the deterministic number D , so even a source observing ready times receives no private coin information from duration. This observation does not enlarge the main theorem into an adaptive-policy optimization result.

Each completed macro update has its rounded stochastic law regardless of padding. Stochastic contraction and a coupling of the at most $N+2$ updates bound final error by $(N+2)(L-1)2^{-b} \leq \delta$. The construction changes no mathematical target to a cutpoint language. $\square$

The comparison/tree construction is standard; the purpose is to price its control and remove the early-stopping side channel noted for the preceding self-paced compiler. It is not zero-error fair-bit sampling of arbitrary irrational probabilities.

Corollary A.2 (Padded label-bit rate). *For every fixed positive tolerance in Theorem 1.1, the exact spherical upper machine has a padded fair-bit approximation with $O(N^{(d-1)/2} \log(N+1))$ labels. In this separately defined nonuniform padded model, minimum persistent label bits obey*

$$\frac{d-1}{2} \log_2 N - O(1) \leq b_N^{\text{pad}} \leq \frac{d-1}{2} \log_2 N + O(\log \log(N+1)).$$

For the five rational gates the leading coefficient is one.

Proof. Lemma 5.2 gives common rows with $L = d+1$ (enlarge L for the binary output if necessary). Compile them with δ equal to the desired fixed tolerance, so $b = O(\log(N+1))$. This proves the upper label and bit bounds. At ready boundaries every padded machine induces an atomic transducer with no more labels than its total control alphabet. The sharp atomic lower bound therefore proves the lower bit bound. Tables, their generation, and arithmetic remain uncharged

in both nonuniform measures. No exact order for the padded label count is inferred from these leading-bit estimates. $\square$

REFERENCES

- [1] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511. Theorem 1; arXiv:1108.6264v1.
- [2] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [3] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics 1730, Springer, 2000.
- [4] M. Iacobelli, Asymptotic quantization for probability measures on Riemannian manifolds, *ESAIM Control Optim. Calc. Var.* **22** (2016), 770–785.
- [5] M. Gu, K. Wiesner, E. Rieper and V. Vedral, Quantum mechanics can reduce the complexity of classical models, *Nature Commun.* **3** (2012), Article 762.
- [6] J. Thompson, A. J. P. Garner, V. Vedral and M. Gu, Using quantum theory to simplify input–output processes, *npj Quantum Inform.* **3** (2017), Article 6.
- [7] F. Ghafari, N. Tischler, J. Thompson, M. Gu, L. K. Shalm, V. B. Verma, S. W. Nam, R. B. Patel, H. M. Wiseman and G. J. Pryde, Dimensional quantum memory advantage in the simulation of stochastic processes, *Phys. Rev. X* **9** (2019), 041013.
- [8] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2 (27 August 2026).
- [9] J. Lumbrellas, H. Ma, J. Thompson and M. Gu, An irreducible quantum advantage in aligning world models with reality, arXiv:2608.19779v1 (2026).
- [10] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *APPROX/RANDOM 2013*, LNCS 8096, 655–670.
- [11] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier-growth bounds for width-3 branching programs, *Theory Comput.* **13** (2017), Article 12, 1–50.
- [12] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs 245, Article 2.
- [13] Q. Qi, General Theta Foundations I: Sharp width of hidden rotation experiments, Revision 35 (2026), publication aabc0731a0c913f7.
- [14] Q. Qi, General Theta Foundations I: Finite alphabets, spectral gaps, and hidden memory, Revision 36 (2026), publication 1a08578710d55a23. Preserved without alteration in the supporting article.